Problem C. Clique Challenge

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	512 megabytes

A clique of a graph G is a set X of vertices of G with the property that every pair of distinct vertices in X are adjacent in G. You are given an undirected graph G with n vertices and m edges, please find the number of distinct non-empty cliques of graph G.

Input

The first line of the input contains two integers n and m ($1 \le n, m \le 1\,000$), denoting the number of vertices and the number of edges.

Each of the following m lines contains two integers u_i and v_i ($1 \le u_i, v_i \le n$, $u_i \ne v_i$), describing an undirected edge between the u_i-th vertex and the v_i-th vertex.

It is guaranteed that there will be at most one edge between each pair of different vertices.

Output

Print a single line containing an integer, denoting the number of cliques. Note that the answer may be extremely large, so please print it modulo $(10^9 + 7)$ instead.

Examples

standard input	standard output
3 2 1 2 2 3	5
3 3 1 2 1 3 2 3	7

Note

In the first example, cliques are $\{1\}$, $\{2\}$, $\{3\}$, $\{1, 2\}$ and $\{2, 3\}$.

In the second example, cliques are $\{1\}$, $\{2\}$, $\{3\}$, $\{1, 2\}$, $\{1, 3\}$, $\{2, 3\}$ and $\{1, 2, 3\}$.